High Availability Web Architecture

AWS Cloud Security Project Report

Course Code: ONL5-ISS11-S5

Course Name: AWS Cloud Security

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GitHub Repository

https://github.com/Rana-Abdelsalamm/AWS-HA-Architecture.git

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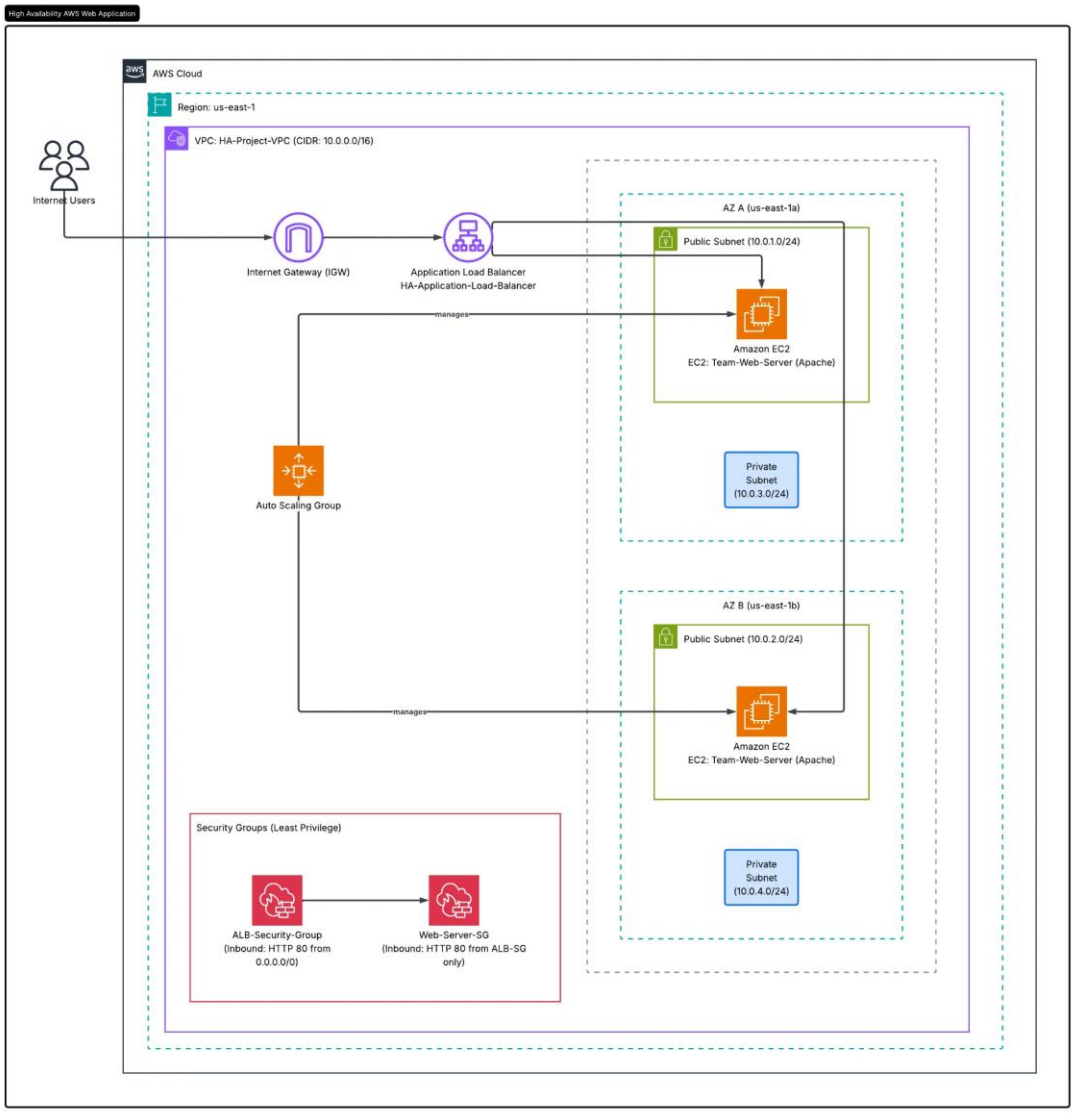
Project Summary and Architecture Diagram

This project implements a highly available and fault-tolerant web architecture on Amazon Web Services using AWS CloudFormation as infrastructure as code. The solution provisions a custom Virtual Private Cloud with IPv4 CIDR block 10.0.0.0/16, public and private subnet tiers distributed across two Availability Zones in the us-east-1 Region, and the routing and security controls required for controlled web access.

Internet users reach an internet-facing Application Load Balancer through an Internet Gateway. The load balancer forwards HTTP traffic on port 80 to Apache web servers managed by an Auto Scaling Group. The group maintains a minimum of two instances and can scale to four, while the target group performs HTTP health checks on the root path. Together, multi-AZ placement, health-based routing, and automated instance replacement reduce the impact of an individual instance or Availability Zone failure and support continuous service availability.

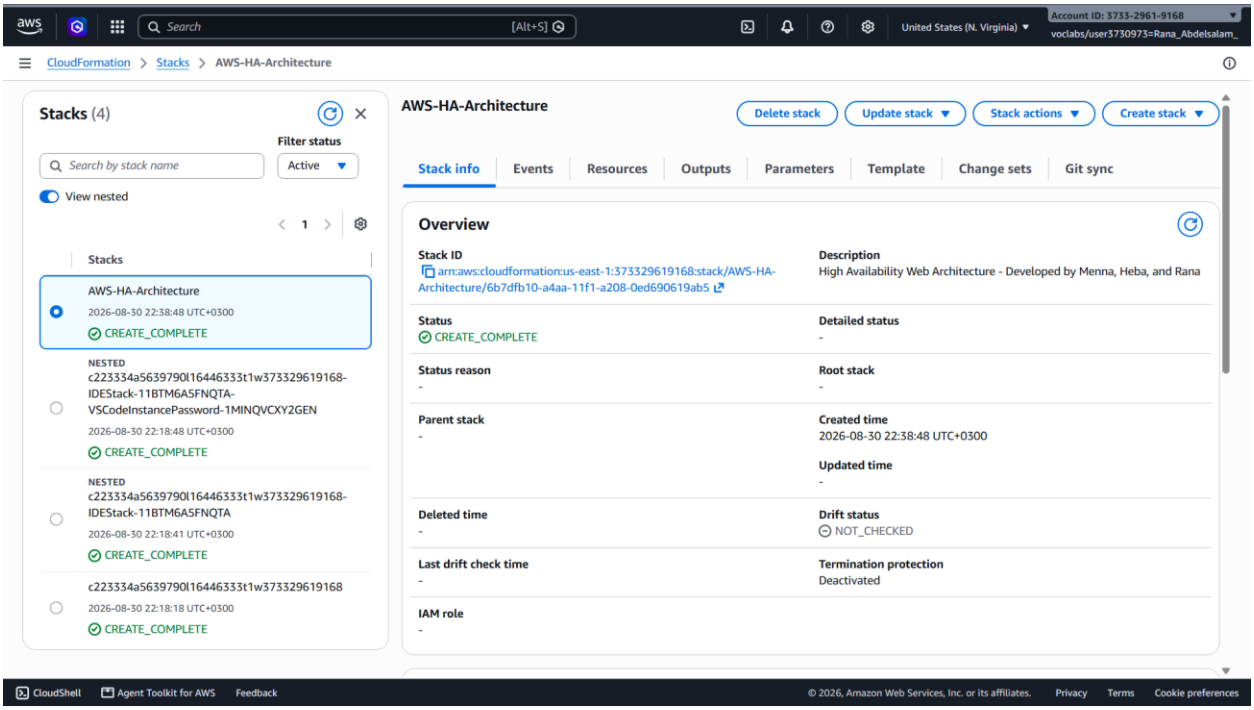


Logical architecture of the AWS high-availability web application.

Implementation Evidence

1. CloudFormation Deployment Completion

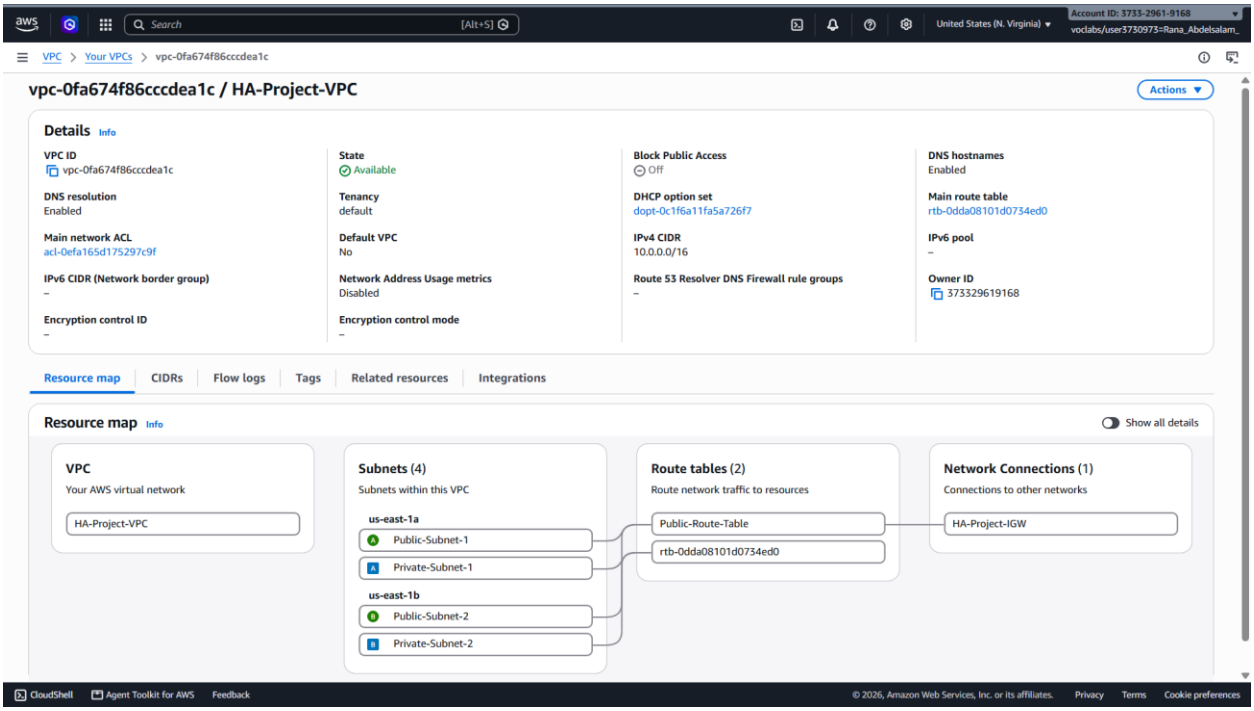
The CloudFormation console confirms that the AWS-HA-Architecture stack was created successfully with status CREATE_COMPLETE. This demonstrates that the infrastructure template was accepted, and its declared AWS resources were provisioned without a deployment failure.



CloudFormation stack in the CREATE_COMPLETE state.

2. VPC and Resource Map

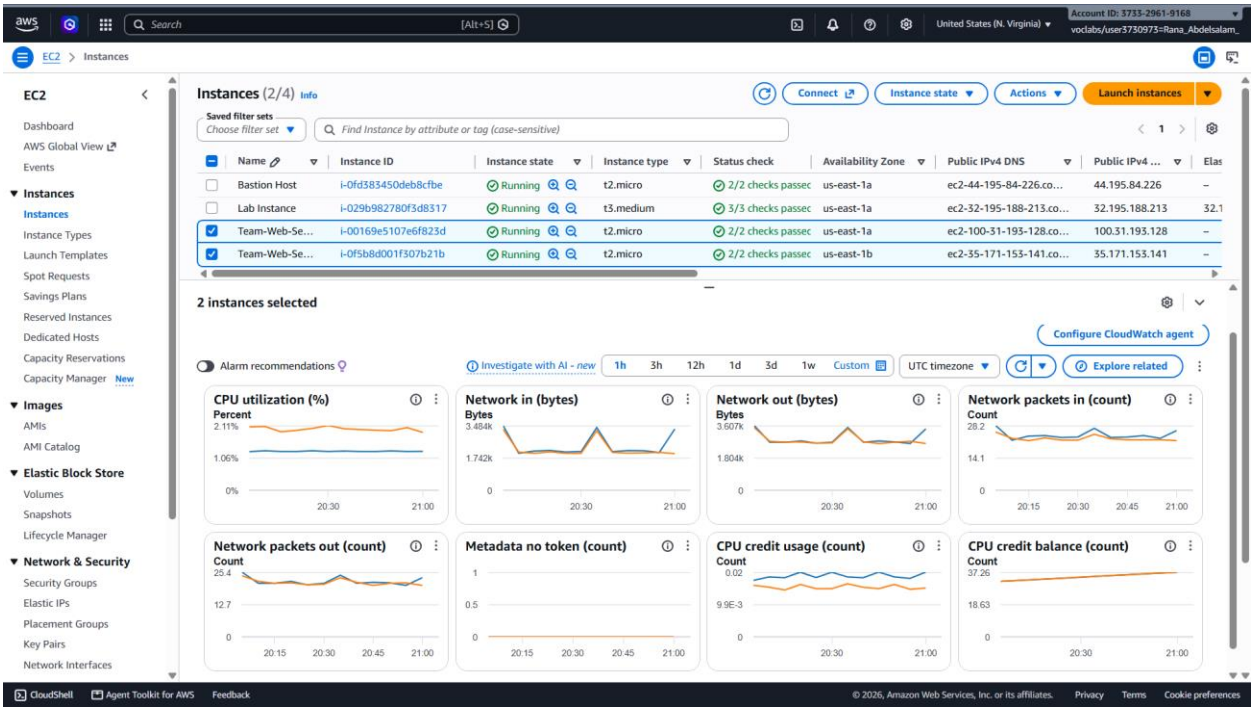
The VPC console shows the deployed HA-Project-VPC as available with CIDR block 10.0.0.0/16, DNS resolution enabled, and DNS hostnames enabled. The resource map confirms four subnets across us-east-1a and us-east-1b, two route tables, and the HA-Project-IGW connection.



Deployed VPC resources distributed across two Availability Zones.

3. EC2 Running Instances

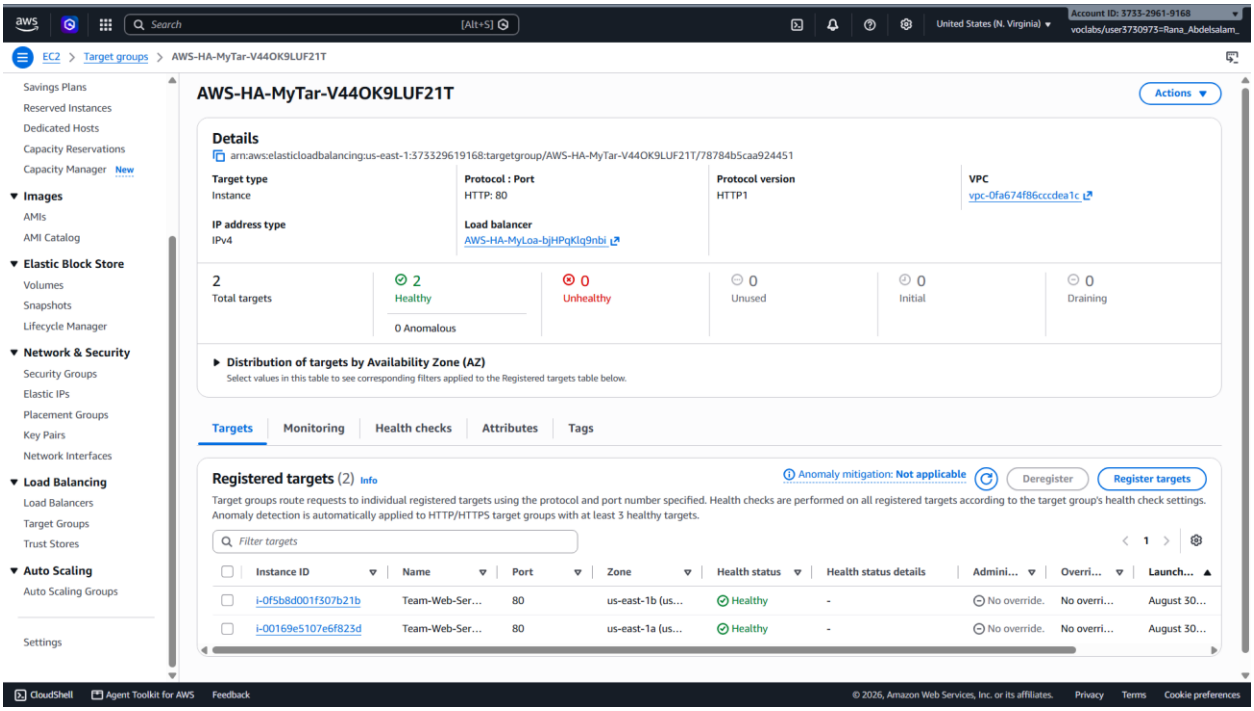
The EC2 console presents two selected Team-Web-Server instances in the Running state. The instances use the t2.micro type, are placed in separate Availability Zones, and report passing status checks, providing runtime evidence of the Auto Scaling Group's baseline capacity.



Two running web-server instances across us-east-1a and us-east-1b.

4. Target Group Health

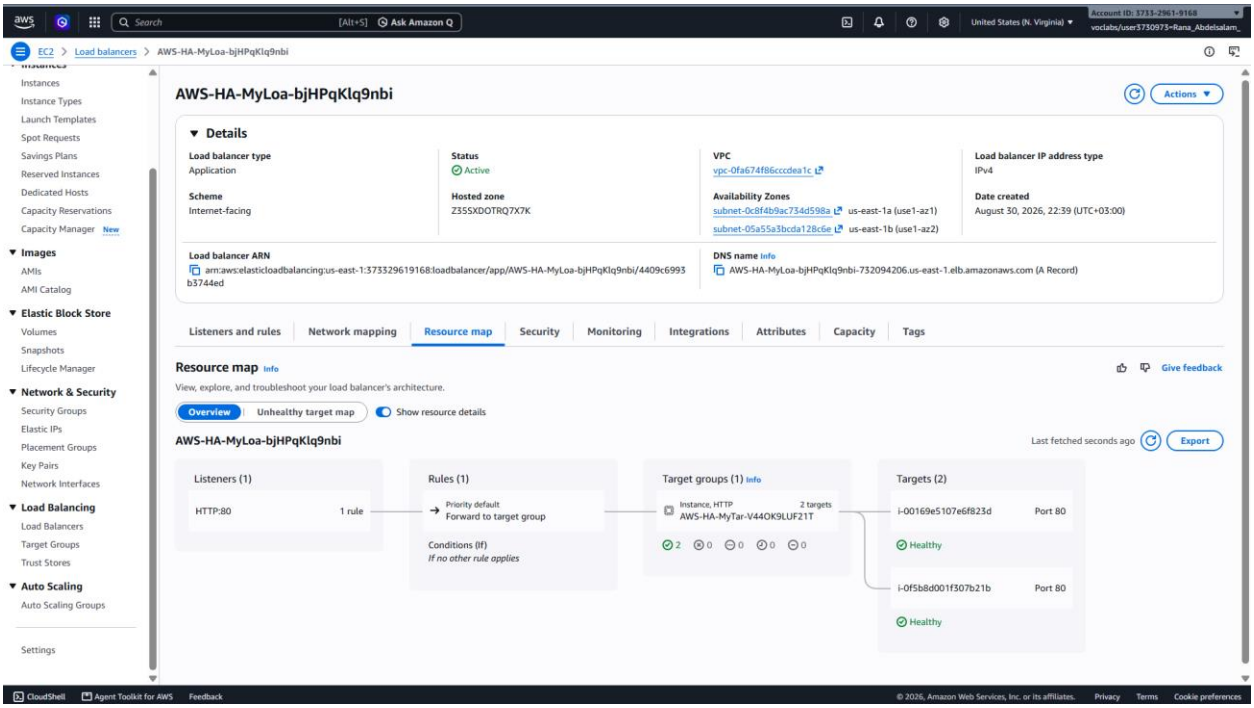
The target group is configured for HTTP traffic on port 80 and reports two total registered targets, with two healthy and zero unhealthy targets. The registered instances are distributed across us-east-1a and us-east-1b, confirming that the load-balancing tier has healthy back-end capacity in both zones.



Target group health status: two healthy targets and zero unhealthy targets.

5. Active Application Load Balancer

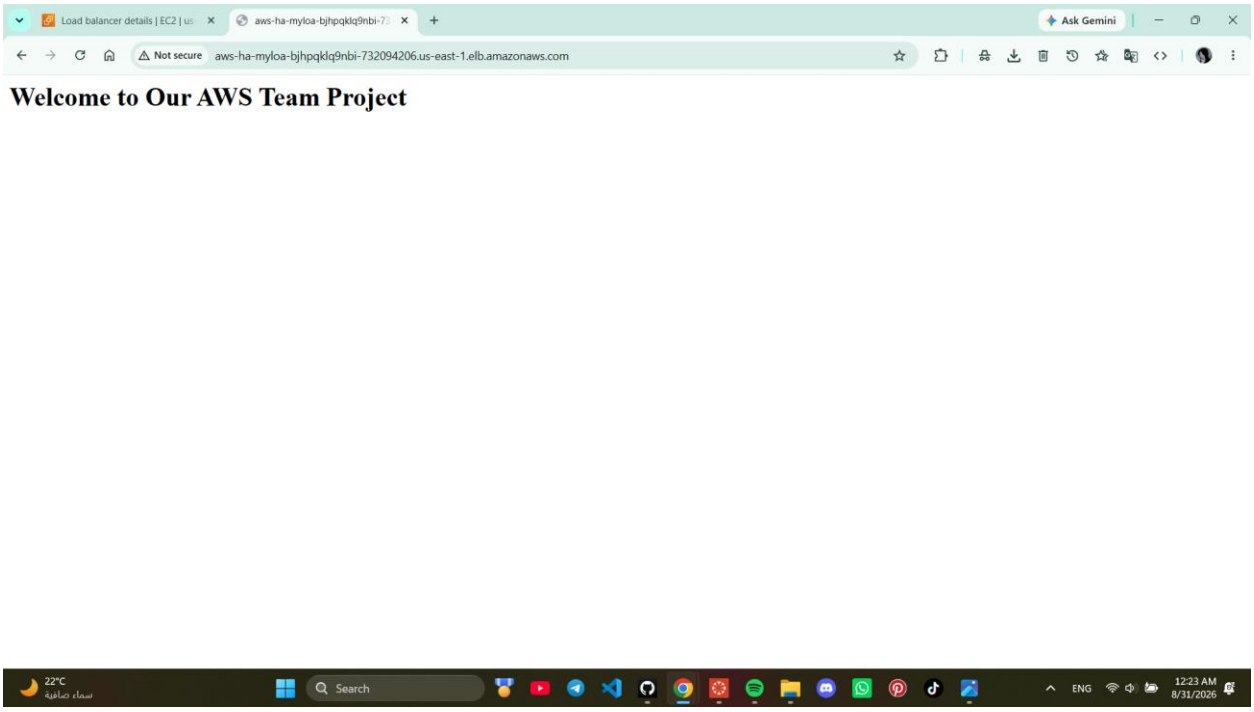
The Load Balancer console shows an internet-facing Application Load Balancer with status Active. Its resource map displays an HTTP:80 listener, forwarding rules to one target group, and two healthy targets across the two configured Availability Zones.



Active internet-facing ALB with HTTP listener and healthy targets.

6. End-to-End Live Web Page

The browser successfully reaches the Application Load Balancer DNS endpoint and renders the Apache-hosted page titled Welcome to Our AWS Team Project. This screenshot is the final end-to-end validation that client traffic can traverse the ALB and reach a healthy web-server target.



Live web page served through the Application Load Balancer endpoint.

CloudFormation Template (Infrastructure as Code)

The following appendix reproduces the complete YAML template used to define the architecture. It includes the custom VPC and subnet topology, Internet Gateway and routing, security groups, EC2 launch template, Application Load Balancer, listener, target group, and Auto Scaling Group.

Code:

```
AWSTemplateFormatVersion: '2010-09-09'
Description: 'High Availability Web Architecture - Developed by Menna, Heba, and
Rana'

Resources:

  # ========================================
  # 1. Menna's Section: VPC & Networking
  # ========================================

  # 1. Main Virtual Private Cloud (VPC)
  VPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: 10.0.0.0/16
      EnableDnsSupport: true
      EnableDnsHostnames: true
      Tags:
        - Key: Name
          Value: HA-Project-VPC

  # 2. Internet Gateway and Attachment
  InternetGateway:
    Type: AWS::EC2::InternetGateway
    Properties:
      Tags:
        - Key: Name
          Value: HA-Project-IGW

  VPCGatewayAttachment:
    Type: AWS::EC2::VPCGatewayAttachment
    Properties:
      VpcId: !Ref VPC
      InternetGatewayId: !Ref InternetGateway

  # 3. Public Subnets (For Load Balancer & Web Servers to access Internet)
  PublicSubnet1:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref VPC
      CidrBlock: 10.0.1.0/24
      AvailabilityZone: us-east-1a
      MapPublicIpOnLaunch: true
      Tags:
        - Key: Name
          Value: Public-Subnet-1
```

```
  PublicSubnet2:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref VPC
      CidrBlock: 10.0.2.0/24
      AvailabilityZone: us-east-1b
      MapPublicIpOnLaunch: true
      Tags:
        - Key: Name
          Value: Public-Subnet-2

  # 4. Private Subnets (Currently not used for servers to allow Apache
installation, but kept for architecture)
  PrivateSubnet1:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref VPC
      CidrBlock: 10.0.3.0/24
      AvailabilityZone: us-east-1a
      MapPublicIpOnLaunch: false
      Tags:
        - Key: Name
          Value: Private-Subnet-1

  PrivateSubnet2:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref VPC
      CidrBlock: 10.0.4.0/24
      AvailabilityZone: us-east-1b
      MapPublicIpOnLaunch: false
      Tags:
        - Key: Name
          Value: Private-Subnet-2

  # 5. Public Route Table and Subnet Associations
  PublicRouteTable:
    Type: AWS::EC2::RouteTable
    Properties:
      VpcId: !Ref VPC
      Tags:
        - Key: Name
          Value: Public-Route-Table

  PublicRoute:
    Type: AWS::EC2::Route
    Properties:
      RouteTableId: !Ref PublicRouteTable
      DestinationCidrBlock: 0.0.0.0/0
      GatewayId: !Ref InternetGateway

  Subnet1Association:
    Type: AWS::EC2::SubnetRouteTableAssociation
```

```
  Properties:
    SubnetId: !Ref PublicSubnet1
    RouteTableId: !Ref PublicRouteTable

Subnet2Association:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref PublicSubnet2
    RouteTableId: !Ref PublicRouteTable

# 6. Security Groups
ALBSecurityGroup:
  Type: AWS::EC2::SecurityGroup
  Properties:
    GroupDescription: Allow HTTP traffic from everywhere to Load Balancer
    VpcId: !Ref VPC
    SecurityGroupIngress:
      - IpProtocol: tcp
        FromPort: 80
        ToPort: 80
        CidrIp: 0.0.0.0/0
    Tags:
      - Key: Name
        Value: ALB-Security-Group

InstanceSecurityGroup:
  Type: AWS::EC2::SecurityGroup
  Properties:
    GroupDescription: Allow HTTP traffic from ALB Security Group only
    VpcId: !Ref VPC
    SecurityGroupIngress:
      - IpProtocol: tcp
        FromPort: 80
        ToPort: 80
        SourceSecurityGroupId: !Ref ALBSecurityGroup
    Tags:
      - Key: Name
        Value: Web-Server-SG


# ========================================
# 2. Heba's Section: EC2 Launch Template
# ========================================

# EC2 Launch Template for the Web Servers
MyLaunchTemplate:
  Type: AWS::EC2::LaunchTemplate
  Properties:
    LaunchTemplateName: WebServerTemplate
    LaunchTemplateData:

      # 1. Amazon Machine Image (Linux) and Instance Type
      ImageId: ami-0a3c3a20c09d6f377
      InstanceType: t2.micro
```

```
        # 2. Security Group Attachment
        SecurityGroupIds:
          - !Ref InstanceSecurityGroup

        # 3. User Data Script to automatically start Apache Web Server
        UserData:
          Fn::Base64: !Sub |
            #!/bin/bash
            yum update -y
            yum install -y httpd
            systemctl start httpd
            systemctl enable httpd
            echo "<h1>Welcome to Our AWS Team Project</h1>" >
/var/www/html/index.html


  # ========================================
  # 3. Rana's Section: ALB & Auto Scaling
  # ========================================

  # 1. Target Group to connect instances with the Load Balancer
  MyTargetGroup:
    Type: AWS::ElasticLoadBalancingV2::TargetGroup
    Properties:
      VpcId: !Ref VPC
      Port: 80
      Protocol: HTTP
      TargetType: instance
      HealthCheckIntervalSeconds: 30
      HealthCheckPath: /
      HealthCheckProtocol: HTTP
      Tags:
        - Key: Name
          Value: HA-Target-Group

  # 2. Application Load Balancer Configuration
  MyLoadBalancer:
    Type: AWS::ElasticLoadBalancingV2::LoadBalancer
    Properties:
      Subnets:
        - !Ref PublicSubnet1
        - !Ref PublicSubnet2
      SecurityGroups:
        - !Ref ALBSecurityGroup
      Scheme: internet-facing
      Tags:
        - Key: Name
          Value: HA-Application-Load-Balancer
```

```
# 3. ALB Listener for incoming traffic
MyListener:
  Type: AWS::ElasticLoadBalancingV2::Listener
  Properties:
    DefaultActions:
      - Type: forward
        TargetGroupArn: !Ref MyTargetGroup
    LoadBalancerArn: !Ref MyLoadBalancer
    Port: 80
    Protocol: HTTP

# 4. Auto Scaling Group Configuration
MyAutoScalingGroup:
  Type: AWS::AutoScaling::AutoScalingGroup
  Properties:
    # Adjusted to Public Subnets so servers can download Apache
    VPCZoneIdentifier:
      - !Ref PublicSubnet1
      - !Ref PublicSubnet2
    LaunchTemplate:
      LaunchTemplateId: !Ref MyLaunchTemplate
      Version: !GetAtt MyLaunchTemplate.LatestVersionNumber
    MinSize: '2'
    MaxSize: '4'
    TargetGroupARNs:
      - !Ref MyTargetGroup
    # Tags added so instances get a clear name when created
    Tags:
      - Key: Name
        Value: Team-Web-Server
        PropagateAtLaunch: true
```